Fixed-Penalty Manifold Augmented Lagrangian Methods

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Outline

Introduction of nonsmooth manifold optimization

A minimax route to fixed penalty

Algorithms and complexity

Numerical tests and conclusions

Problem

Consider the following nonsmooth problem on manifold:

$$\boxed{\min_{x \in \mathcal{M}} \ \varphi(x) := f(x) + h(\mathcal{A}x)}$$

Standing assumptions.

- $\mathcal{M} \subset \mathbb{R}^n$ is a compact embedded submanifold;
- f is smooth and may be nonconvex; $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ is linear; h is convex, Lipschitz continuous, and possibly nonsmooth.

Applications.

- sparse principal component analysis;
- robust rotation averaging and synchronization;
- large-model training, including parameter-efficient methods such as *GaLore* and *LoRA*.

Nonsmooth optimization on manifolds

- **Subgradient and gradient-sampling methods:** extend subgradient methods from Euclidean convex or weakly convex to Riemannian geodesically convex or weakly convex objectives, while gradient sampling handles locally Lipschitz, generally nonconvex objectives using several nearby gradients or convex-hull subproblems (Ferreira and Oliveira, 1998; de Carvalho Bento et al., 2016; Li et al., 2021; Hosseini and Uschmajew, 2017; Hosseini et al., 2018).

- **Composite proximal and Newton-type methods:** exploit $f(x) + h(x)$ through first-order models on the tangent space and, when available, using semismooth/Newton methods as inner solvers. (Chen et al., 2020; Huang and Wei, 2022, 2023; Si et al., 2024).

- **Smoothing and continuation:** smoothing the nonsmooth-term $h(\mathcal{A}x)$ with a vanishing smoothing parameter (Peng et al., 2023; Beck and Rosset, 2023; Zhang et al., 2021).

- **Splitting and multiplier methods:** introduce the multiplier and design agorithms based on Lagrangian or augmented Lagrangian functions. (Lai and Osher, 2014; Kovnatsky et al., 2016; Chen et al., 2016; Deng and Peng, 2022; Li et al., 2024; Deng et al., 2025a; Xu et al., 2025; Yuan, 2024; Deng et al., 2025b).

Manifold augmented Lagrangian method

- Introduce $y = \mathcal{A}x$:

$$\min_{x\in\mathcal{M},\, y\in\mathbb{R}^m} f(x) + h(y) \quad \text{s.t.} \quad \mathcal{A}x - y = 0.$$

  Eliminating y directly defines the reduced augmented Lagrangian:

$$\begin{aligned}\mathcal{L}_\sigma(x,\lambda) &= \min_y \left\{ f(x) + h(y) - \langle \lambda, \mathcal{A}x - y\rangle + \frac{\sigma}{2}\|\mathcal{A}x - y\|^2 \right\} \\ &= f(x) + M_h^{1/\sigma}(\mathcal{A}x - \lambda/\sigma) - \frac{1}{2\sigma}\|\lambda\|^2.\end{aligned}$$

- Introducing $y_\sigma(x,\lambda) = \operatorname{prox}_{h/\sigma}(\mathcal{A}x - \lambda/\sigma)$, a basic inexact manifold ALM (Deng and Peng, 2022; Zhou et al., 2023) is:

$$\begin{cases} x_{k+1} \approx \arg\min\limits_{x\in\mathcal{M}} \mathcal{L}_{\sigma_k}(x, \lambda_k), \\ \lambda_{k+1} = \lambda_k - \sigma_k\big(\mathcal{A}x_{k+1} - y_\sigma(x_{k+1}, \lambda_k)\big). \end{cases}$$

The x-subproblem is smooth on $\mathcal{M}$, although the original objective contains the nonsmooth term h.

A brief history of ALM theory on manifolds

1. **Early splitting schemes without convergence.** Directly applying ADMM to manifold setting works well numerically, but no theorectical guarantte. (Lai and Osher, 2014; Kovnatsky et al., 2016).

2. **General manifold ALM.** Manifold ALMs (Deng and Peng, 2022; Zhou et al., 2023) have asymptotic convergence guarantee for nonsmooth composite problems on embedded submanifolds with careful control on the subproblem solving. **Penalty parameters are increasing and can be unbounded.**

3. **Non-asymptotic theory.** ManIAL and RiAL reach $\mathcal{O}(\epsilon^{-3})$ first-order oracle complexity by a careful design of penalty parameters and dual stepsizes (Deng et al., 2025a; Xu et al., 2025). **Penalty parameters are increasing and unbounded.**

All existing ALM methods for nonsmooth optimization on manifolds require the penalty parameter to tend to infinity.

Existing Manifold ALMs: Double Loops and Increasing Penalties

All existing manifold ALMs are double-loop, and require increasing penalties.

Method	Inner solver for x-subproblem	Guarantee	Penalty
MIALM (Deng and Peng, 2022)	gradient descent	asymptotic convergence	increasing
SSN-based ALM (Zhou et al., 2023)	semismooth Newton (SSN)	asymptotic / local fast convergence	increasing
ManIAL / RiAL (Deng et al., 2025a; Xu et al., 2025)	Riemannian first-order methods	$\mathcal{O}(\epsilon^{-3})$	increasing

Can we design a single-loop ALM with one fixed penalty parameter σ?

Why a fixed penalty is meaningful

For the AL function, the Moreau-envelope term contributes a smoothness scale of order

$$L_x(\sigma) \;\sim\; L_f + \sigma\|\mathcal{A}\|^2 \quad \text{(up to geometry/retraction constants)}.$$

Hence growing σ may force increasingly ill-conditioned subproblems (given that the smallest eigenvalue may be unchanged).

- **Constraint-aware interpretation:** σ controls how strongly the model reacts to constraint violation; a reasonably large fixed σ is therefore meaningful in practice.
- **Closer to practice:** safeguarded updates often increase σ only when constraint violation is unsatisfactory, then hold it once progress is adequate.
- **Consistent with convex theory:** in convex optimization, convergence with a fixed penalty parameter is standard; this motivates asking when the same principle survives here.

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A minimax reformulation

The KKT conditions of the original composite problem are

$$\boxed{\exists\, g \in \partial h(\mathcal{A}x): \qquad P_{T_x\mathcal{M}}\big(\nabla f(x) + \mathcal{A}^* g\big) = 0.}$$

For any fixed $\sigma > 0$, consider

$$\boxed{\min_{x\in\mathcal{M}} \max_{\lambda\in\mathbb{R}^m} \mathcal{L}_\sigma(x,\lambda).}$$

- Dual stationarity is feasibility: $\nabla_\lambda \mathcal{L}_\sigma(x,\lambda) = y_\sigma(x,\lambda) - \mathcal{A}x = 0$.
- Together with proximal optimality, it gives $y_\sigma(x,\lambda) = \mathcal{A}x$ and $g := -\lambda \in \partial h(y_\sigma(x,\lambda))$.
- Primal stationarity then becomes $P_{T_x\mathcal{M}}(\nabla f(x) - \mathcal{A}^*\lambda) = 0$.
- Hence, for *every* fixed $\sigma > 0$,

$$\mathrm{grad}_x\, \mathcal{L}_\sigma(x,\lambda) = 0, \quad \nabla_\lambda \mathcal{L}_\sigma(x,\lambda) = 0 \iff (x, y_\sigma(x,\lambda), \lambda) \text{ is a split KKT point.}$$

Intuitively, maximizing in λ enforces feasibility, while minimizing in x enforces tangential stationarity; the saddle-point conditions are exactly the original KKT conditions.

Minimax optimization: what theory is available?

- **Euclidean nonconvex–concave methods:** two-time-scale GDA, smoothed GDA, alternating gradient projection, and proximal-point schemes; in the merely concave setting, compactness of the maximization domain is typically required (Lin et al., 2020; Xu et al., 2023; Rafique et al., 2022; Zhang et al., 2020).

  Without compactness, additional regularity conditions, such as one-sided Polyak–Łojasiewicz growth, are needed (Yang et al., 2022).

- **Geodesically convex–concave minimax on manifolds:** saddle-point and extragradient frameworks exploit compatible curvature and geodesic convexity–concavity (Jordan et al., 2022; Zhang et al., 2023; Cai et al., 2023; Hu et al., 2024).

- **Nonconvex–concave minimax on manifolds:** current results cover strongly concave or linear maximization, and nonsmooth models over compact dual sets (Huang and Gao, 2023; Xu et al., 2024b,a; Aybat et al., 2025).

How can we get rid of compactness in the nonconvex–merely-concave ALM saddle-point problem?

Contributions

We add a vanishing dual regularizer to improve the concavity, as in minimax literature (Xu et al., 2023):

$$F_k(x,\lambda) = \mathcal{L}_\sigma(x,\lambda) - \frac{\mu_k}{2}\|\lambda\|^2, \qquad \mu_k \downarrow 0,$$

and exploit the special AL structure to obtain:

- **Single loop:** one multiplier step and one Riemannian gradient step on x per iteration.
- **Fixed penalty parameter:** σ is independent of both k and the target accuracy.
- **No compactness requirement:** the multiplier iterates are controlled by the model itself, without projecting onto a compact dual set.
- **Best-known guarantee:** $\mathcal{O}(\epsilon^{-4})$ oracle complexity, matching the best-known order for single-loop nonconvex–merely-concave minimax methods.

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GDA based on fixed-penalty AL saddle point problem

At iteration k, regularize only the multiplier side:

$$F_k(x, \lambda) = \mathcal{L}_\sigma(x, \lambda) - \frac{\mu_k}{2}\|\lambda\|^2.$$

Then perform single-step first-order updates on both λ and x:

$$\begin{cases} \lambda_{k+1} = \lambda_k + \sigma \nabla_\lambda F_k(x_k, \lambda_k), \\ x_{k+1} = \mathrm{Retr}_{x_k}(-\alpha_k \operatorname{grad}_x \mathcal{L}_\sigma(x_k, \lambda_{k+1})). \end{cases}$$

Gradient descent-ascent iteration for $\min_x \max_\lambda F_k(x, \lambda)$!

- Introducing $y_\sigma(x_k, \lambda_k) = \mathrm{prox}_{h/\sigma}(\mathcal{A}x_k - \lambda_k/\sigma)$, the multiplier step is

$$\lambda_{k+1} = (1 - \sigma\mu_k)\lambda_k - \sigma(\mathcal{A}x_k - y_\sigma(x_k, \lambda_k)),$$

- Per-iteration cost: One vector proximal evaluation, one Riemannian-gradient step, no inner loop, and one fixed σ.

Convergence: $\mathcal{O}(\epsilon^{-4})$ iteration complexity

Theorem (Single-loop $\mathcal{O}(\epsilon^{-4})$ complexity)

Choose

$$\mu_k = c_\mu k^{-1/4}, \qquad \alpha_k = c_\alpha \mu_k^2, \qquad \sigma c_\mu \le \tfrac{1}{2},$$

where c_α is sufficiently small. Then there exists a tail iterate $\tilde{k}$ such that

$$\| \operatorname{grad}_x \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1}) \| + \| \nabla_\lambda \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1}) \| = \mathcal{O}(K^{-1/4}).$$

Consequently, the corresponding split triple is an ϵ-KKT point after

$$K = \mathcal{O}(\epsilon^{-4}).$$

The method is genuinely single-loop: every iteration uses one vector proximal evaluation and one Riemannian-gradient update.

Proof roadmap

- **Step 1: bound the multipliers.** Use the Moreau-envelope gradient bound and the damped multiplier recursion; no compact dual set is imposed.
- **Step 2: obtain primal descent through a value function.** Maximize out the regularized multiplier, apply Danskin's theorem, and isolate the multiplier drift error.
- **Step 3: control the drift error of the multipliers.** Combine contraction of one multiplier step with the drifts caused by moving x_k and decreasing μ_k.

Step 1: The multipliers are uniformly bounded

Let $v_k = \mathcal{A}x_k - \lambda_k/\sigma$. Since h is ℓ_h-Lipschitz, the Moreau-envelope gradient is bounded:

$$\sigma\|v_k - \operatorname{prox}_{h/\sigma}(v_k)\| = \|\nabla M_h^{1/\sigma}(v_k)\| \le \ell_h.$$

Substituting $\mathcal{A}x_k = v_k + \lambda_k/\sigma$ into the damped multiplier update gives

$$\lambda_{k+1} = \sigma\big(\operatorname{prox}_{h/\sigma}(v_k) - v_k\big) - \sigma\mu_k\lambda_k.$$

Taking norms and using the Moreau-gradient bound yields the scalar recursion

$$\begin{aligned}\|\lambda_{k+1}\| &\le \sigma\|\operatorname{prox}_{h/\sigma}(v_k) - v_k\| + \sigma\mu_k\|\lambda_k\| \\ &\le \ell_h + \sigma\mu_k\|\lambda_k\|.\end{aligned}$$

If $\sigma\mu_k \le q < 1$, iteration yields

$$\|\lambda_k\| \le \max\left\{\|\lambda_1\|, \frac{\ell_h}{1-q}\right\}; \qquad q \le \tfrac{1}{2} \implies \|\lambda_k\| \le \max\{\|\lambda_1\|, 2\ell_h\}.$$

Boundedness comes from the Moreau-envelope structure—there is no projection onto a compact set.

Step 2: Value function and primal descent

Define the exact regularized maximizer and its value function:

$$\lambda_k^\star(x) := \arg\max_\lambda F_k(x,\lambda), \qquad \psi_k(x) := F_k(x, \lambda_k^\star(x)).$$

Strong concavity makes $\lambda_k^\star(x)$ unique, and Danskin's theorem gives

$$\operatorname{grad} \psi_k(x) = \operatorname{grad}_x \mathcal{L}_\sigma(x, \lambda_k^\star(x)).$$

Retraction smoothness of ψ_k and the computed primal step imply

$$\psi_{k+1}(x_{k+1}) \le \psi_k(x_k) - \frac{\alpha_k}{4}\|\operatorname{grad}\psi_k(x_k)\|^2 + \frac{L_{x\lambda}^2 \alpha_k}{2}\|\lambda_{k+1} - \lambda_k^\star(x_k)\|^2 + \frac{\mu_k - \mu_{k+1}}{2}\|\lambda_{k+1}^\star(x_{k+1})\|^2.$$

Primal descent is obtained up to a multiplier tracking error and the drift from $\mu_k \downarrow 0$.

Step 3: Drift error recrusion

Set $e_k = \|\lambda_k - \lambda_k^\star(x_k)\|^2$. One ascent step with length σ contracts toward the *current* maximizer:

$$\|\lambda_{k+1} - \lambda_k^\star(x_k)\|^2 \le (1 - \sigma\mu_k) e_k.$$

But the target moves because both x_k and μ_k change. Controlling these two drifts gives

$$e_{k+1} \le \left(1 - \frac{\sigma\mu_k}{2}\right) e_k + C_1 \frac{\alpha_k^2}{\mu_k^3} \left(\|\operatorname{grad} \psi_k(x_k)\|^2 + e_k\right) + C_2 \frac{1}{\mu_k} \left(\frac{\mu_k - \mu_{k+1}}{\mu_{k+1}}\right)^2.$$

- x-drift uses $\|\lambda_k^\star(x) - \lambda_k^\star(x')\| \le (L_{\lambda x}/\mu_k)\|x - x'\|$.
- μ-drift uses strong concavity and the uniform bound on $\lambda_k^\star(x)$.

Putting the three steps together

Choose $\mu_k = c_\mu k^{-1/4}$ and $\alpha_k = c_\alpha \mu_k^2$, and couple primal descent with tracking through

$$V_k := \psi_k(x_k) - \underline{\psi} + \theta_0 \mu_k e_k.$$

For sufficiently small c_α, the tracking contraction absorbs the $+\alpha_k e_k$ term in primal descent, giving

$$\sum_{k=1}^{K} \alpha_k \| \operatorname{grad} \psi_k(x_k) \|^2 + \sum_{k=1}^{K} \mu_k^2 e_k \le C.$$

Since both weights sum to $\Theta(K^{1/2})$ over the last half of the iterates, some tail index $K/2 \le \tilde{k} \le K$ satisfies

$$\| \operatorname{grad} \psi_{\tilde{k}}(x_{\tilde{k}}) \| + \| \lambda_{\tilde{k}+1} - \lambda^\star_{\tilde{k}}(x_{\tilde{k}}) \| = \mathcal{O}(K^{-1/4}).$$

Finally,

$$\| \operatorname{grad}_x \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1}) \| \le \underbrace{\| \operatorname{grad} \psi_{\tilde{k}}(x_{\tilde{k}}) \|}_{\mathcal{O}(K^{-1/4})} + L_{x\lambda} \underbrace{\| \lambda_{\tilde{k}+1} - \lambda^\star_{\tilde{k}}(x_{\tilde{k}}) \|}_{\mathcal{O}(K^{-1/4})} = \mathcal{O}(K^{-1/4}),$$

$$\| \nabla_\lambda \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1}) \| \le (\sigma^{-1} + \mu_k) \| \lambda_{k+1} - \lambda^\star_k \| + \underbrace{\mu_k \| \lambda_{k+1} \|}_{\mathcal{O}(\mu_k) \text{ by Step 1}}.$$

Thus the KKT residual is $\mathcal{O}(K^{-1/4})$, hence $K = \mathcal{O}(\epsilon^{-4})$. **This matches the best-known complexity.**

Can we match the nonsmooth manifold benchmark?

- **Single-loop contribution.** A fixed-penalty method with no compact dual-domain assumption and $\mathcal{O}(\epsilon^{-4})$ KKT complexity.
- **The complexity gap.** The $\mathcal{O}(\epsilon^{-4})$ rate does not match the $\mathcal{O}(\epsilon^{-3})$ benchmark for nonsmooth manifold optimization.
- **Observation.** One multiplier-ascent step leaves a drift error that must be coupled with primal descent, forcing the conservative choice of primal step size $\alpha_k = \Theta(\mu_k^2)$.
- **Our double-loop strategy.** Control the multiplier residual more accurately, while still taking only one primal Riemannian-gradient step per outer iteration.

Fixed-penalty double-loop ALM

At outer iteration k, define the dually regularized AL function

$$F_k(x, \lambda) = \mathcal{L}_\sigma(x, \lambda) - \frac{\mu_k}{2}\|\lambda\|^2, \qquad \mu_k \downarrow 0,$$

while keeping the augmented-Lagrangian penalty $\sigma > 0$ fixed.

1. **Inner multiplier loop:** approximately solve

$$\max_{\lambda \in \mathbb{R}^m} F_k(x_k, \lambda) \quad \text{until} \quad \|\nabla_\lambda F_k(x_k, \lambda_{k+1})\| \le \varepsilon_k.$$

2. **One primal step:**

$$x_{k+1} = \mathrm{Retr}_{x_k}(-\alpha_k \,\mathrm{grad}_x\, \mathcal{L}_\sigma(x_k, \lambda_{k+1}))\,.$$

Solving the multiplier subproblem

Let $d_k = \mathcal{A}x_k$ and

$$\nabla_\lambda F_k(x_k, \lambda) = \operatorname{prox}_{h/\sigma}(d_k - \lambda/\sigma) - d_k - \mu_k \lambda, \qquad L_{\lambda,k} = \frac{1}{\sigma} + \mu_k.$$

General proximable h

Use gradient ascent:

$$\lambda^{t+1} = \lambda^t + \frac{1}{L_{\lambda,k}} \nabla_\lambda F_k(x_k, \lambda^t),$$

until

$$\|\nabla_\lambda F_k(x_k, \lambda^t)\| \le \varepsilon_k.$$

Separable $h(u) = \sum_i h_i(u_i)$

The optimality condition decouples into

$$q_{i,k}(\lambda_i) := \operatorname{prox}_{h_i/\sigma}(d_{i,k} - \lambda_i/\sigma) - \mu_k \lambda_i - d_{i,k} = 0.$$

Each $q_{i,k}$ is continuous and strictly decreasing, so solve it by bisection on $[-\ell_i, \ell_i]$ until

$$|q_{i,k}(\lambda_i)| \le \varepsilon_k / \sqrt{m}.$$

The two solvers use no evaluations of ∇f, no tangent projections, and no retractions inside the multiplier loop.

Double-loop complexity

Theorem (Fixed-penalty double-loop complexity)

Let $\mu_k = c_\mu k^{-1/3}$, $\alpha_k = c_\alpha k^{-1/3}$, *and* $\varepsilon_k = k^{-2/3}$. *Then a tail iterate is* $\mathcal{O}(K^{-1/3})$*-KKT. Consequently,*

$$\boxed{N_{\text{primal}} = \mathcal{O}(\epsilon^{-3})}$$

with fixed σ. *Moreover,*

$$N_{\text{multiplier}} = \begin{cases} \widetilde{\mathcal{O}}(\epsilon^{-4}) & \text{generic gradient ascent / vector prox,} \\ \widetilde{\mathcal{O}}(m\epsilon^{-3}) & \text{coordinate bisection for separable } h. \end{cases}$$

- **Primal and multiplier oracles.** A primal oracle uses ∇f, $\mathcal{A}^*$, tangent projection, and retraction; a multiplier oracle uses one vector proximal evaluation or scalar root operations.

The multiplier oracle is inexpensive.

Multiplier boundedness in the double-loop method

For fixed x, let $\lambda_k^\star(x) = \arg\max_\lambda F_k(x,\lambda)$. Its first-order condition and the Moreau-gradient bound imply

$$0 = \nabla_\lambda F_k(x, \lambda_k^\star(x)) \quad \Longrightarrow \quad \left(\mu_k + \frac{1}{\sigma}\right) \|\lambda_k^\star(x)\| \le \frac{\ell_h}{\sigma},$$

and therefore

$$\boxed{\|\lambda_k^\star(x)\| \le \frac{\ell_h}{1+\sigma\mu_k}.}$$

For the computed multiplier, the inner stopping condition gives

$$\|\nabla_\lambda F_k(x_k, \lambda_{k+1})\| \le \varepsilon_k \quad \Longrightarrow \quad \boxed{\|\lambda_{k+1}\| \le \frac{\ell_h + \sigma\varepsilon_k}{1+\sigma\mu_k}.}$$

Both multipliers are controlled by the AL structure and the inner residual—not by projection onto a compact set.

Improved drift control

The inner stopping condition and μ_k-strong concavity give the direct tracking bound

$$\|\lambda_{k+1} - \lambda_k^\star(x_k)\| \le \frac{\|\nabla_\lambda F_k(x_k, \lambda_{k+1})\|}{\mu_k} \le \frac{\varepsilon_k}{\mu_k}.$$

Set the value-function to be potential function, the following descent inequality:

$$\psi_{k+1}(x_{k+1}) \le \psi_k(x_k) - \frac{\alpha_k}{4}\|\operatorname{grad}\psi_k(x_k)\|^2 + \frac{L_{x\lambda}^2 \alpha_k \varepsilon_k^2}{2\mu_k^2} + \frac{\mu_k - \mu_{k+1}}{2}\|\lambda_{k+1}^\star(x_{k+1})\|^2.$$

With $\mu_k, \alpha_k = \Theta(k^{-1/3})$ and $\varepsilon_k = k^{-2/3}$, the two error terms are summable over a tail, while $\sum_{k=\lceil K/2\rceil}^{K} \alpha_k = \Theta(K^{2/3})$. Hence some $k \le \tilde{k} \le K$ satisfies

$$\|\operatorname{grad}\psi_{\tilde{k}}(x_{\tilde{k}})\| = \mathcal{O}(K^{-1/3}).$$

Finally,

$$\|\operatorname{grad}_x \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1})\| \le \|\operatorname{grad}\psi_{\tilde{k}}\| + L_{x\lambda}\frac{\varepsilon_{\tilde{k}}}{\mu_{\tilde{k}}}, \qquad \|\nabla_\lambda \mathcal{L}_\sigma(x_{\tilde{k}}, \lambda_{\tilde{k}+1})\| \le \varepsilon_{\tilde{k}} + \mu_{\tilde{k}}\|\lambda_{\tilde{k}+1}\|,$$

so multiplier boundedness and $\varepsilon_k/\mu_k = \Theta(k^{-1/3})$ give an $\mathcal{O}(K^{-1/3})$ KKT residual.

Comparison with existing manifold ALMs

Method	Inner subproblem	Guarantee	Penalty
MIALM	inexact smooth manifold solve	asymptotic convergence	increasing
SSN-based ALM	semismooth Newton (SSN)	asymptotic / local fast convergence	increasing
ManIAL / RiAL	iterative Riemannian first-order solve	$\mathcal{O}(\epsilon^{-3})$	increasing
Single-loop (ours)	one Riemannian-gradient step	$\mathcal{O}(\epsilon^{-4})$	fixed
Double-loop (ours)	inexact multiplier solver	$\mathcal{O}(\epsilon^{-3})$	fixed

Our methods keep the primal update explicit and the penalty fixed; the double-loop variant also matches the best-known $\mathcal{O}(\epsilon^{-3})$ complexity.

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Experimental results

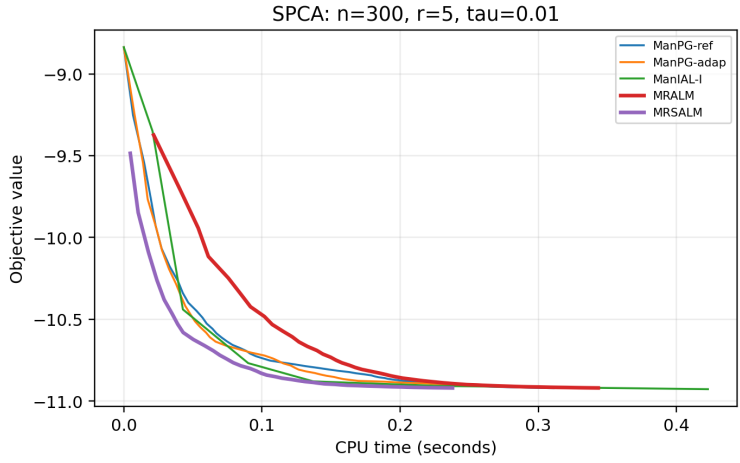
- Problems: sparse PCA and compressed modes over Stiefel manifolds.

$$\text{Sparse PCA:} \quad \min_{X \in \mathrm{St}_{n,r}} -\operatorname{tr}(X^\top C^\top C X) + \tau \|X\|_1,$$

$$\text{Compressed modes:} \quad \min_{X \in \mathrm{St}_{n,r}} \operatorname{tr}(X^\top H X) + \mu \|X\|_1.$$

- Baselines: ManPG, adaptive ManPG, ManIAL-I, and a Riemannian subgradient method.
- Proposed methods: double-loop MRALM and single-loop MRSALM with fixed parameters across each instance grid.
- Same initialization, random seed, and objective-target stopping rule; ManPG supplies the reference objective.

Sparse PCA: fixed parameters across the grid



The fixed-penalty variants reach the common objective target reliably and are competitive in elapsed time, without a growing primal penalty schedule.

Compressed modes: time to a common objective target

n	r	μ	ManPG	ManPG-adap	ManIAL-I	MRALM	MRSALM / speedup
256	10	.01	13.55	6.01	1.37	2.47	**0.72** / $18.82\times$
256	20	.01	0.98	0.70	0.46	0.58	**0.25** / $3.92\times$
512	10	.01	17.28	11.24	**1.08**	6.82	2.23 / $7.75\times$
512	20	.01	5.97	3.75	2.16	4.76	**2.07** / $2.88\times$
256	10	.02	18.06	8.90	–	4.21	**1.20** / $15.05\times$
256	20	.02	6.21	3.86	–	8.83	**2.65** / $2.34\times$
512	10	.02	20.86	9.86	–	9.29	**2.99** / $6.98\times$
512	20	.02	8.33	4.81	–	10.74	**3.65** / $2.28\times$

Times are in seconds; "–" means that the common objective target was not reached.

MRSALM reaches the target on all 8 instances, is fastest on 7, and achieves a median $5.45\times$ speedup over ManPG.

Conclusions

- **Single loop.** One multiplier update and one Riemannian-gradient step per iteration, with no inner subproblem solve; the method achieves $\mathcal{O}(\epsilon^{-4})$ KKT complexity.
- **Fixed penalty parameter.** The penalty σ remains fixed throughout the algorithm and is independent of the target accuracy.
- **No compact dual-domain assumption.** Multiplier boundedness follows from dual regularization and the augmented-Lagrangian structure, without projecting onto a prescribed compact set.
- **Sharper double-loop variant.** More accurate multiplier solves recover the best-known $\mathcal{O}(\epsilon^{-3})$ primal-oracle complexity while retaining fixed σ.

Can a single-loop method attain $\mathcal{O}(\epsilon^{-3})$ complexity?

- **A possible route: moving-center regularization.** Inspired by the smoothing idea of Zhang et al. (2020), consider

$$\max_{\lambda \in \mathbb{R}^m} \left\{ \mathcal{L}_\sigma(x, \lambda) - \frac{\mu}{2} \|\lambda - \zeta\|^2 \right\}.$$

- **Why might this improve the rate?** If ζ tracks λ, the KKT bias may be small without weakening the dual regularization too aggressively.

- **The remaining challenge.** How should ζ and μ be updated, and can the AL structure control both multiplier tracking and moving-center drift in a single loop? Whether this yields $\mathcal{O}(\epsilon^{-3})$ is still unclear.

Thank you!

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